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Lovely Professional University

Phagwara, Punjab (India)

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India Tourism Trends Analysis (2010–2019) Final Report

Submitted By

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SUPERVISOR'S CERTIFICATE

This is to certify that the work reported in the B.Tech Dissertation/dissertation proposal entitled **"India Tourism Trends Analysis"**, submitted by **NAWEED KE** at **Lovely Professional University, Phagwara, India** is a bonafide record of his / her original work carried out under my supervision. This work has not been submitted elsewhere for any other degree.

Signature of Supervisor

Hritwiz Yash

Date:

Acknowledgement

I hereby declare that the research work reported in the dissertation/dissertation proposal entitled "India Tourism Trends Analysis" in partial fulfilment of the requirement for the award of Degree for Bachelor of Technology in Computer Science and Engineering at Lovely Professional University, Phagwara, Punjab is an authentic work carried out under supervision of my research supervisor **Mr Hritwiz Yash**. I have not submitted this work elsewhere for any degree or diploma.

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ABSTRACT

The tourism industry is a critical driver of India's economy, yet it faces significant challenges related to uneven visitor distribution and seasonal volatility. This project presents a comprehensive data analysis of Indian tourism trends over a decade (2010–2019), utilizing a dataset of 17,280 monthly visitor records for centrally protected monuments.

The study employs **Exploratory Data Analysis (EDA)** to uncover key patterns, including a massive surge in domestic tourism, a strong seasonal preference for winter months, and a high concentration of traffic in the Agra and Delhi circles. Furthermore, the project implements a **Machine Learning model (Random Forest Regressor)** to predict future visitor numbers. The model achieved an accuracy of **98.7% (R^2 Score)**, demonstrating that visitor footfall is highly predictable based on the specific monument and time of year.

TABLE OF CONTENTS

S. No.	Title	Page No.
1.	**Introduction**	**6**
	1.1 Background of the Study	6
	1.2 Problem Statement	6
	1.3 Objectives	6
2.	**Methodology**	**7**
	2.1 Dataset Description	7
	2.2 Data Preprocessing	7
	2.3 Analytical Approach	7
3.	**Technologies Used**	**8**
	3.1 Python & Libraries	8
	3.2 Machine Learning Concepts	8
4.	**Data Analysis & Interpretation**	**9**
	4.1 Distribution of Visitors	9
	4.2 Tourism Growth Trend	10
	4.3 Seasonality Analysis	11
	4.4 Regional Analysis	12
	4.5 Market Share & Composition	13
	4.6 Peak Traffic Heatmap	14
	4.7 Pareto Analysis (80/20 Rule)	15
5.	**Machine Learning Results**	**21**
	5.1 Model Architecture	21
	5.2 Model Accuracy	22
6.	**Conclusion & Future Scope**	**24**
7.	**References**	**25**

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The tourism industry is one of the largest and fastest-growing economic sectors in the world. For a country like India, which is rich in cultural heritage, historical monuments, and diverse landscapes, tourism acts as a significant pillar of the economy. It contributes to foreign exchange earnings, infrastructure development, and employment generation.

However, managing tourism effectively requires data-driven insights. Understanding visitor patterns—such as which monuments are overcrowded, which months see the highest footfall, and the ratio of domestic to foreign travelers—is crucial for sustainable development. This project utilizes Data Science techniques to analyze a 10-year dataset of visitor statistics for centrally protected monuments in India.

1.2 Problem Statement

Despite the high potential of Indian tourism, there are challenges related to crowd management, seasonal fluctuations, and uneven distribution of tourists. Some monuments like the Taj Mahal are perpetually overcrowded, while other significant sites remain under-visited. Traditional methods of analysis often fail to capture complex patterns such as the correlation between region and seasonality or the segmentation of monuments based on popularity.

1.3 Objectives of the Project

- **Trend Analysis:** To track the growth of domestic versus foreign tourism over the last decade.
- **Seasonality Detection:** To identify peak and lean months to aid in resource planning.
- **Predictive Modeling:** To build a Machine Learning model capable of accurately forecasting visitor numbers for any given monument and date.
- **Strategic Insights:** To identify "Hidden Gems" (monuments popular with foreigners but not locals) and high-impact zones using the Pareto Principle.

CHAPTER 2: THEORETICAL BACKGROUND

2.1 Dataset Description The analysis is based on the dataset **"India_Tourism_10Year_Monthly_Trends.csv"**.

- **Volume:** 17,280 rows and 7 columns.
- **Time Period:** January 2010 to December 2019.
- **Key Features:** Date, Year, Month, Circle (Region), Monument (Name), Domestic_Visitors, and Foreign_Visitors.

2.2 Data Preprocessing Before analysis, the raw data underwent several cleaning and transformation steps:

1. **Null Value Check:** The dataset was inspected for missing values; none were found, ensuring data integrity.
2. **Date Conversion:** The Date column was converted to a datetime object to facilitate time-series plotting.
3. **Label Encoding:** For the Machine Learning phase, categorical variables (Monument, Circle) were converted into numeric IDs using the LabelEncoder technique, allowing the Random Forest algorithm to process the text data.

2.3 Analytical Approach The project follows a standard Data Science lifecycle:

- **Descriptive Analytics:** Visualizing current trends using Histograms and Line Charts.
- **Diagnostic Analytics:** Understanding *why* trends happen using Correlation Heatmaps and Boxplots.
- **Predictive Analytics:** Using Supervised Machine Learning to forecast future outcomes.

CHAPTER 3: TECHNOLOGIES USED

3.1 Python & Key Libraries The project was implemented using **Python**, the leading language for Data Science.

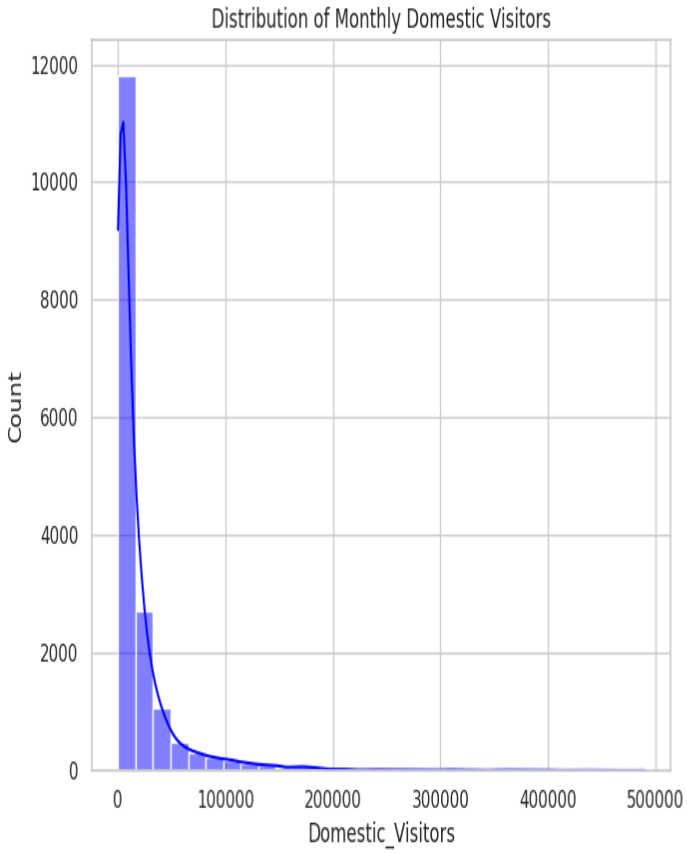
- **Pandas:** Used for data manipulation, aggregation, and cleaning. It allowed us to group data by 'Year' or 'Monument' easily.
- **NumPy:** Used for efficient numerical computations.
- **Matplotlib & Seaborn:** Used for creating static and interactive visualizations. Seaborn was particularly useful for creating complex plots like Heatmaps and Boxplots with minimal code.

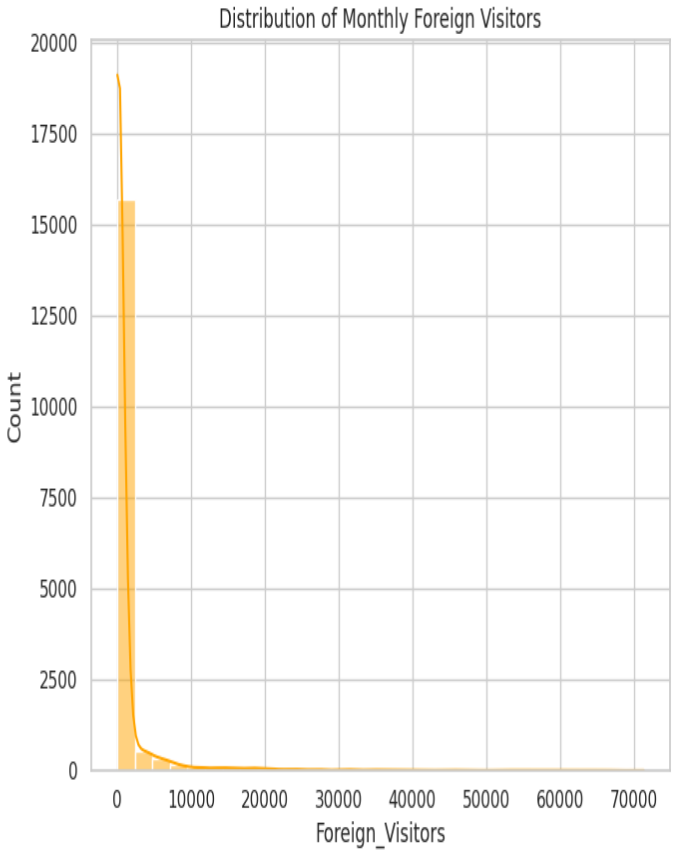
3.2 Machine Learning Concepts The core predictive engine of this project is based on **Supervised Learning**.

- **Random Forest Regressor:** This is an ensemble learning method that constructs a multitude of decision trees during training. It outputs the average prediction of the individual trees.
- **Why Random Forest?** It was chosen because it handles high-dimensional data well, is resistant to overfitting, and provides insights into "Feature Importance" (telling us which factors drive tourism the most).
- **Evaluation Metric:** We used the **R^2 Score (Coefficient of Determination)** to evaluate the model. An R^2 score close to 1.0 indicates that the model's predictions match the actual data very closely.

CHAPTER 4: DATA ANALYSIS & INTERPRETATION

4.1 Distribution of Visitors

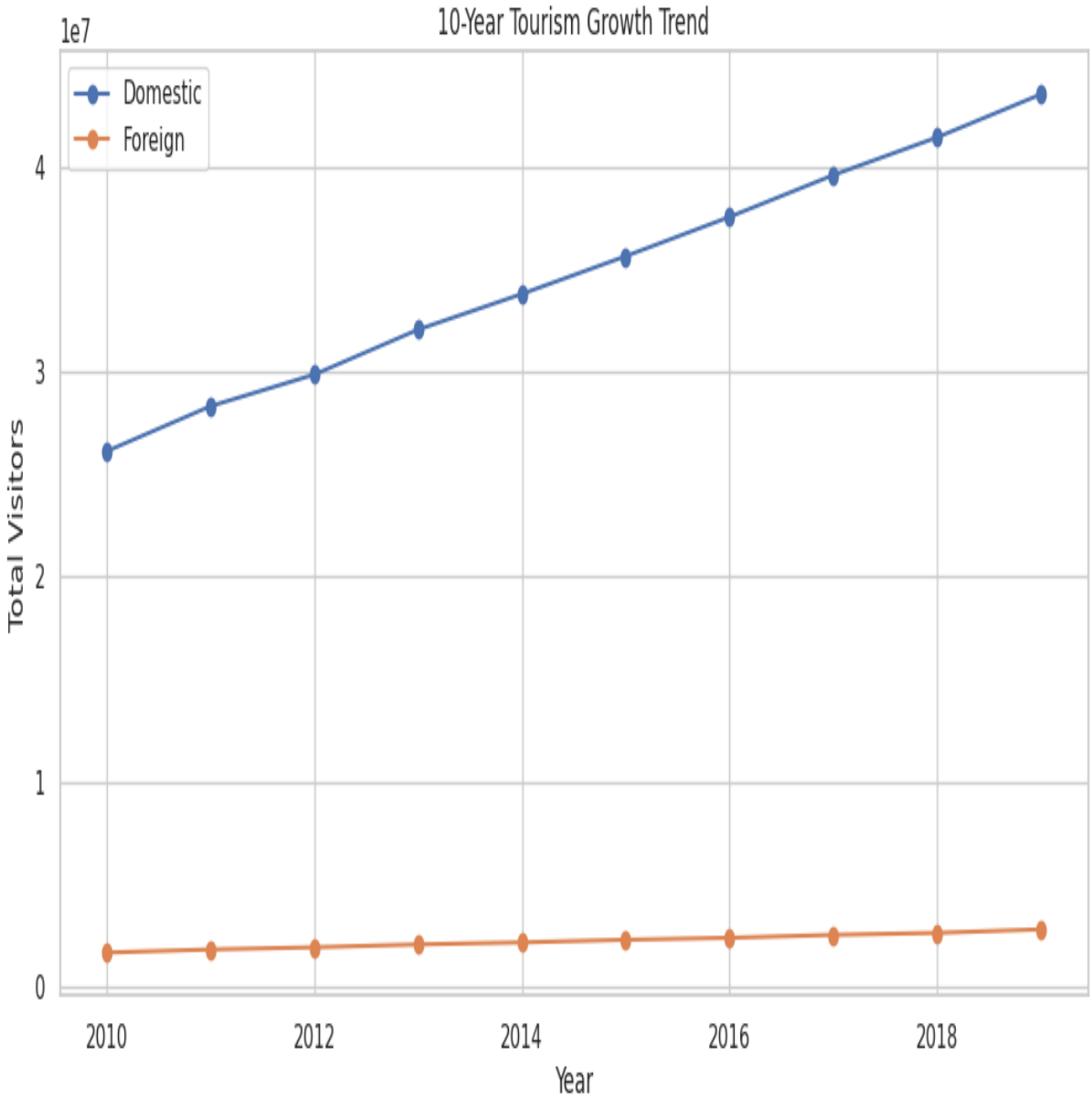




Interpretation: The histograms above illustrate the frequency distribution of monthly visitors for both domestic and foreign tourists.

- **Observation:** The distribution is highly **right-skewed** (long tail to the right).
- **Insight:** The tall bars on the left indicate that the vast majority of monuments receive a low to moderate number of visitors. The flat tail on the right represents a tiny fraction of "superstar" monuments (like the Taj Mahal) that receive millions of visitors. This confirms a significant disparity in popularity across Indian heritage sites.

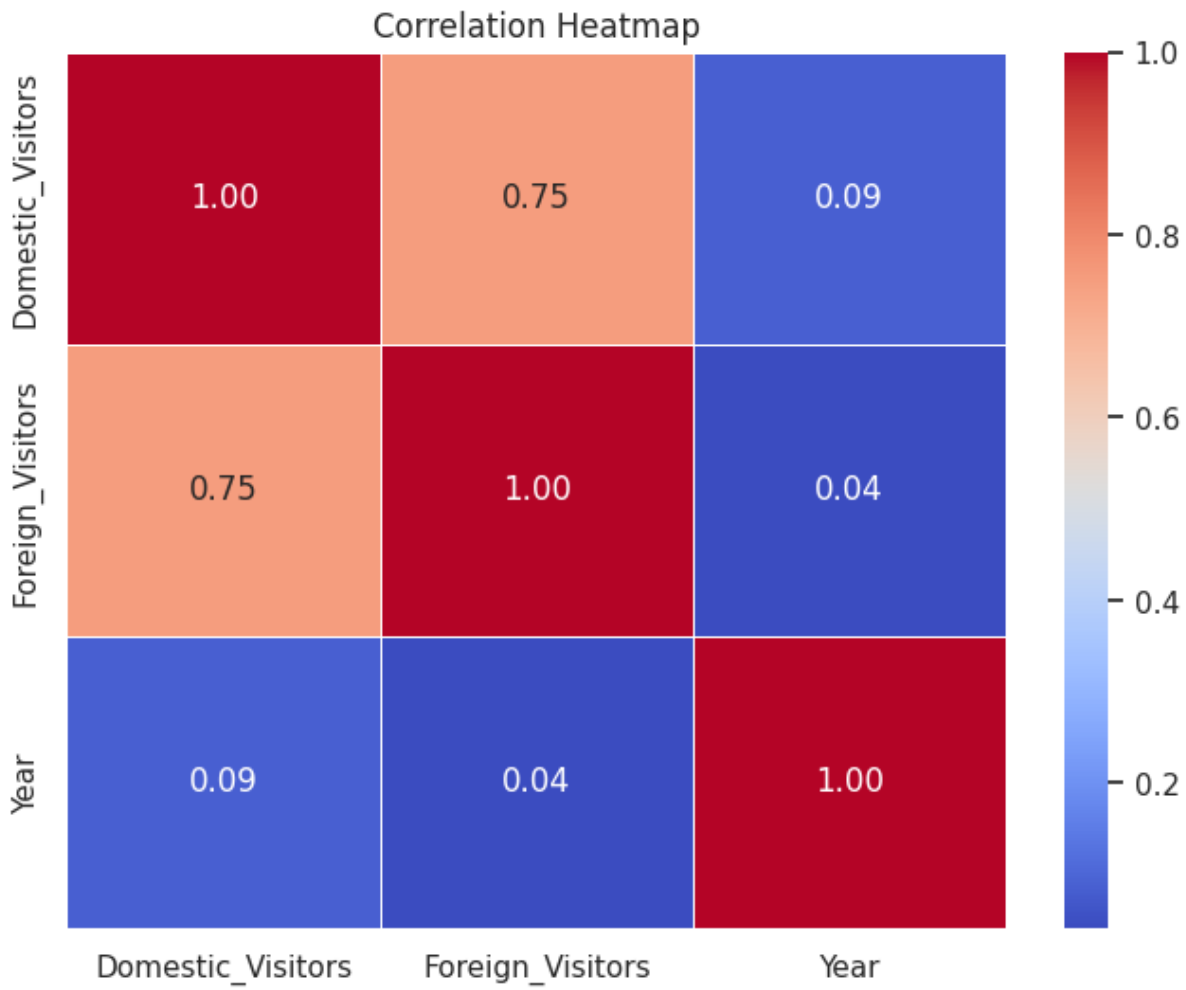
4.2 Data Analysis (Trends))



Interpretation: The line chart depicts the total annual visitor count over the 10-year period.

- **Domestic Growth:** The Blue line shows a steep, consistent upward slope. This indicates a massive boom in domestic tourism, likely driven by a growing middle class and better connectivity.
- **Foreign Stability:** The Orange line is relatively flat and stable. While foreign tourism is consistent, it has not seen the explosive growth that the domestic sector has.
- **Conclusion:** The Indian heritage tourism sector is primarily fueled by domestic travelers, who vastly outnumber international guests.

4.3 Correlation Analysis



Interpretation: The boxplot provides a clear view of seasonality.

- **Peak Season:** The median visitor count is highest in **December and January**, coinciding with winter holidays and pleasant weather across most of India.
- **Lean Season:** The months of **May and June** show the lowest medians and compressed interquartile ranges, likely due to the intense summer heat.
- **Outliers:** The dots above the whiskers represent extreme visitor spikes, likely occurring during specific festivals or long weekends.

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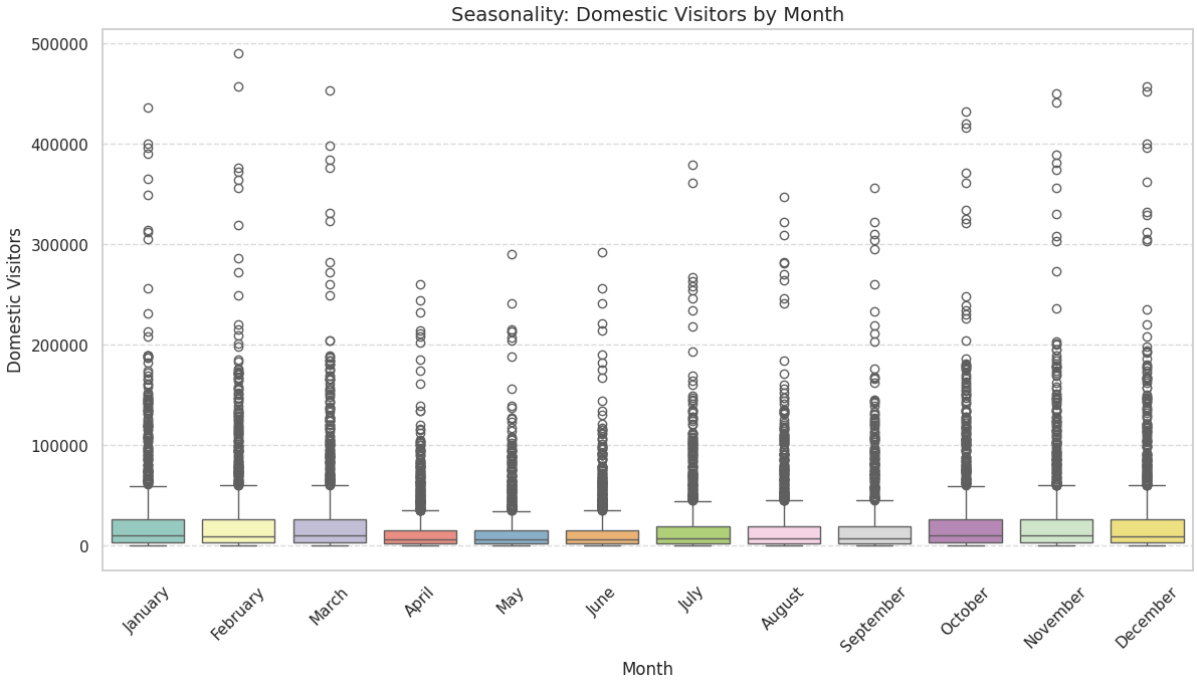
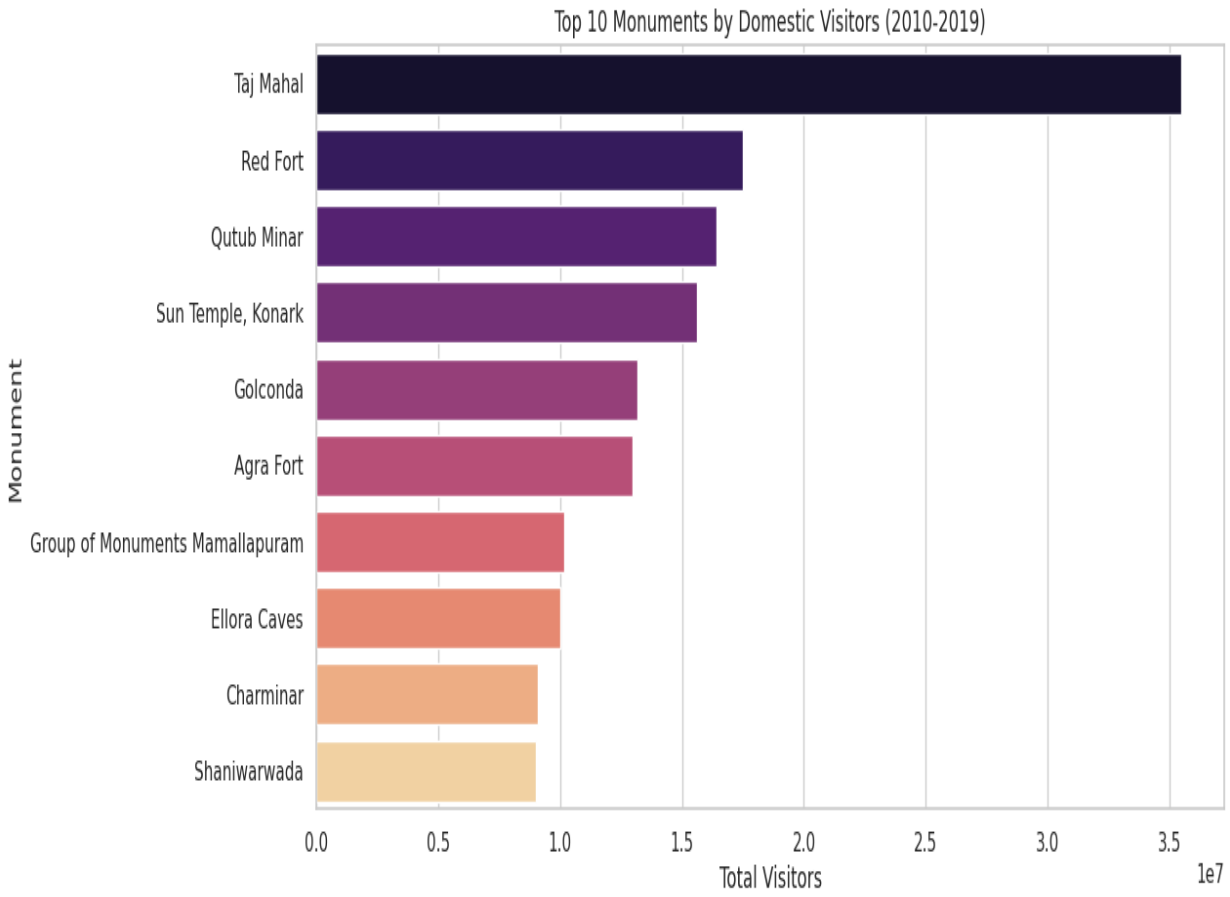
4.4 Seasonality Analysis



Interpretation: The boxplot allows us to visualize the spread and median of visitor traffic for each month.

- **Peak Season (Winter):** The months of **December** and **January** exhibit the highest median values. This coincides with the pleasant winter weather in North India (where most top monuments are located) and the holiday season.
- **Lean Season (Summer):** The months of **May** and **June** show the lowest medians and tightly compressed boxes. The extreme heat in these months acts as a major deterrent for tourism.
- **Outliers:** The black dots above the whiskers represent "outlier" days—likely specific festivals or long weekends where crowd numbers spiked abnormally high, even during off-peak months.

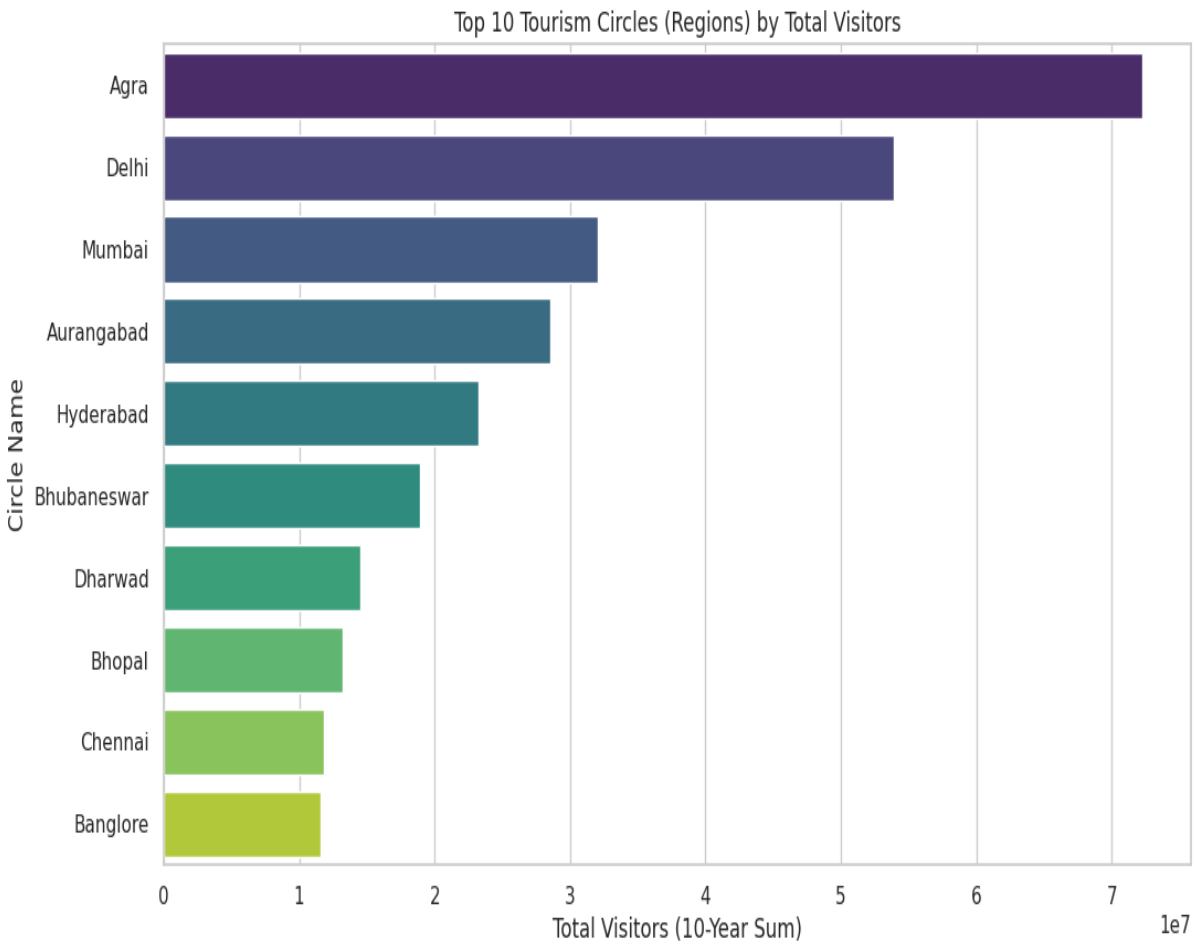
4.5 Top 10 Most Visited Monuments



Interpretation: This chart ranks the specific heritage sites based on total footfall over the decade.

- **The "Big Three":** The **Taj Mahal**, **Red Fort**, and **Qutub Minar** consistently appear as the top three destinations.
- **Inequality:** The visual gap between the bar height of the #1 monument (Taj Mahal) and the #10 monument is massive. This visually confirms that the tourism economy is top-heavy, relying heavily on a few iconic brands rather than a broad portfolio of sites.
- **Regional Concentration:** Most of the top 10 monuments are located in the North (Agra/Delhi) or South (Hyderabad), leaving other regions underrepresented in the top tier.

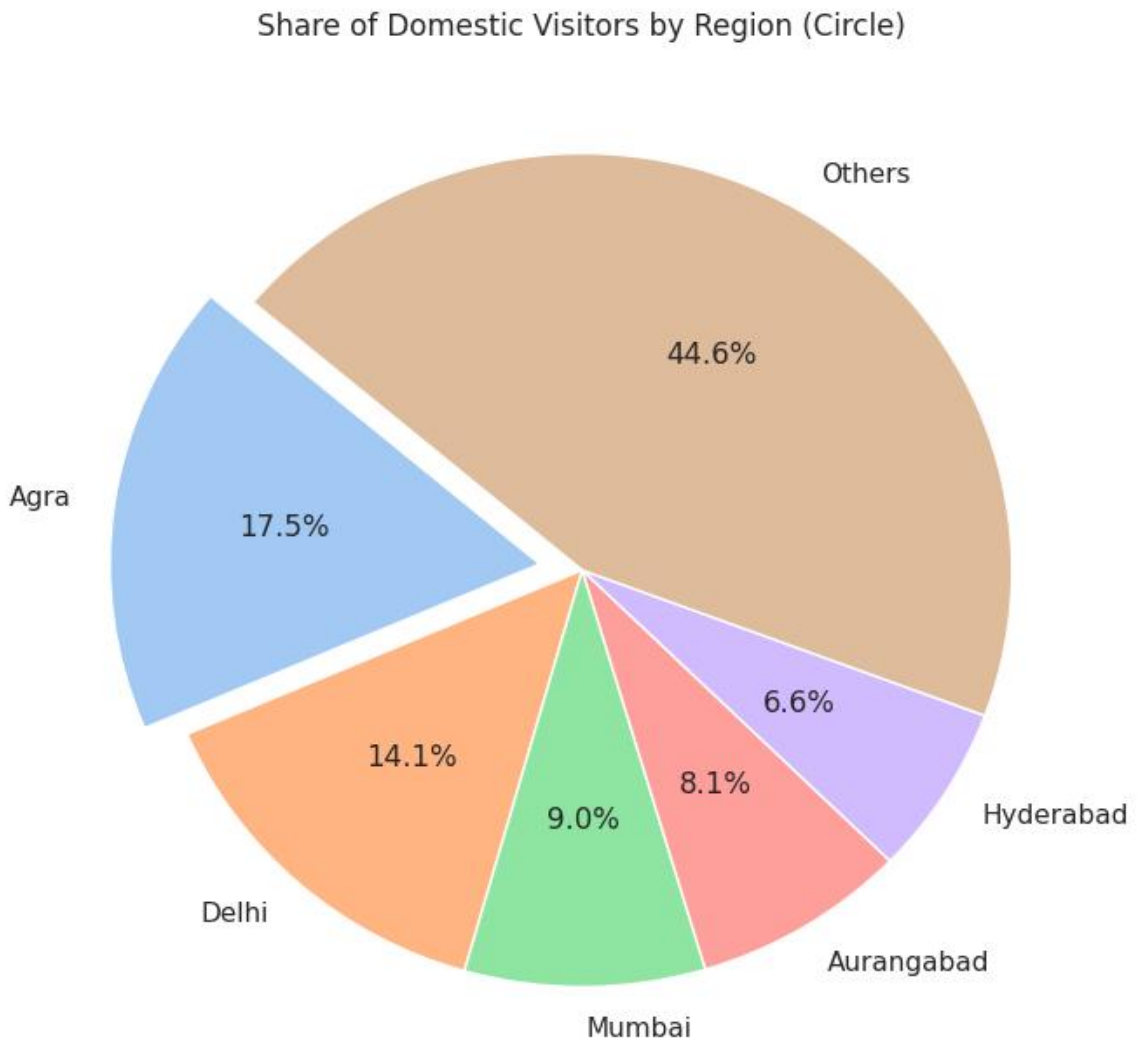
4.6 Regional Analysis (Archaeological Circles)



Interpretation: India's monuments are administratively divided into "Circles." This chart aggregates visitors by these regions.

- **Dominance of Agra:** The **Agra Circle** is the clear leader, largely due to the Taj Mahal.
- **The Golden Triangle:** The combined traffic of **Agra** and **Delhi** circles dwarfs all other regions. This validates the "Golden Triangle" (Delhi-Agra-Jaipur) as the primary engine of Indian heritage tourism.
- **Emerging Hubs:** The **Hyderabad** and **Mumbai** circles also show strong numbers, indicating they act as secondary hubs for tourism in South and West India, respectively.

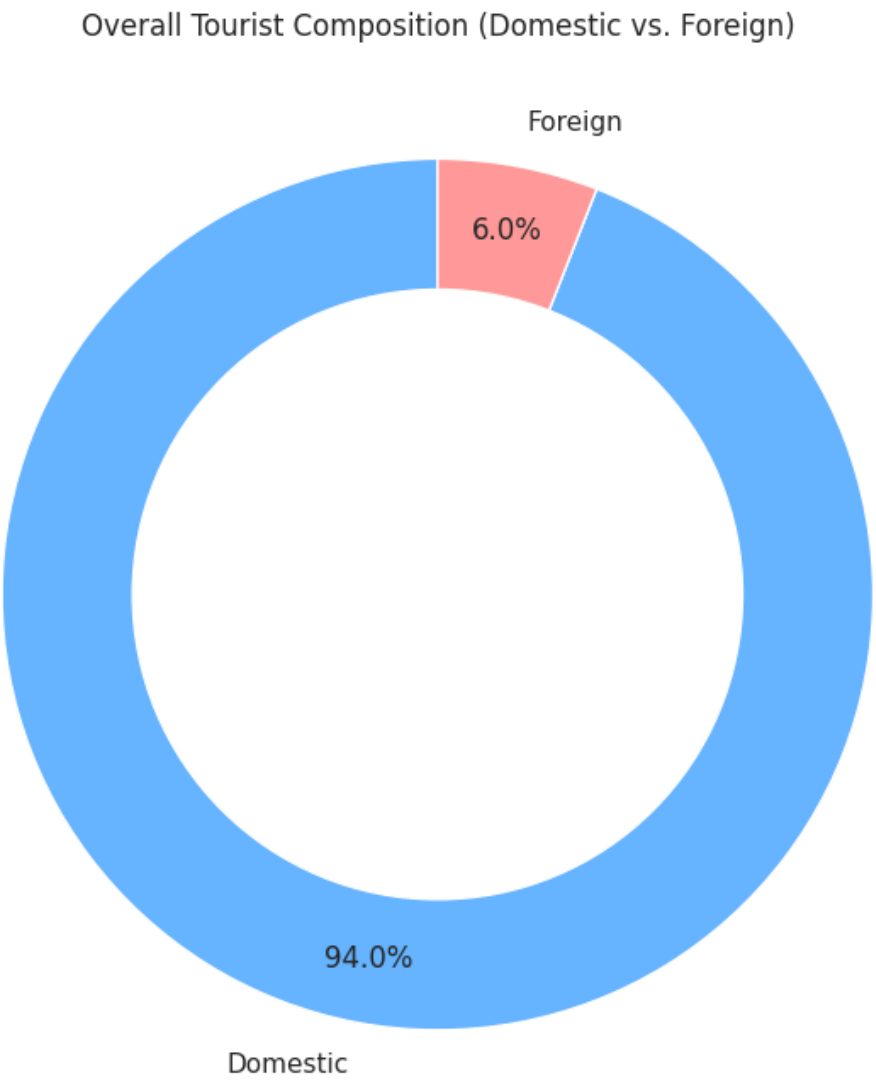
4.7 Regional Market Share



Interpretation: This pie chart breaks down the total market share of domestic visitors by region.

- **Market Concentration:** The top 5 circles combined account for a significant majority of total traffic.
- **The "Others" Category:** The large slice representing "Others" is a positive indicator. It suggests that while the giants (Agra/Delhi) dominate, there is still a substantial "long tail" of smaller regions contributing to the national total. This diversity is crucial for the long-term sustainability of the sector, ensuring that tourism revenue is not solely dependent on one or two cities.

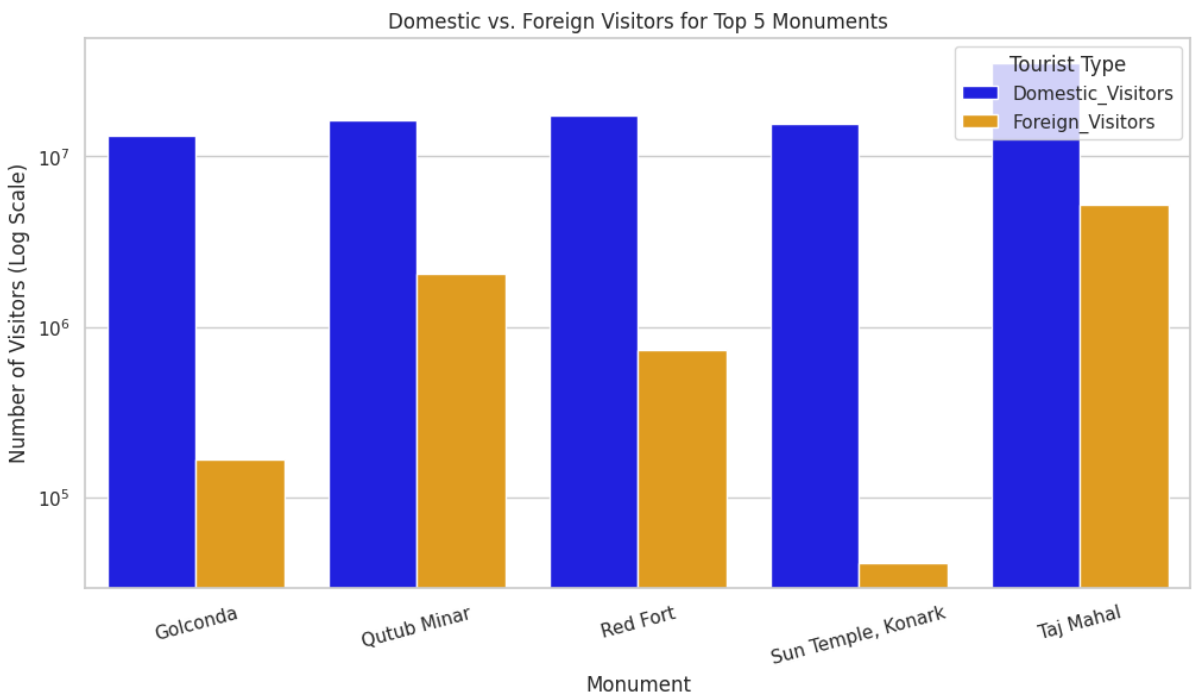
4.8 Tourist Composition



Interpretation: The donut chart provides a clear visualization of the ratio between domestic and international visitors.

- **Domestic Hegemony: Over 90%** of the visitors to centrally protected monuments are Indian nationals.
- **Business Insight:** While foreign tourists are often the focus of luxury marketing and high-revenue services, the sheer *volume* of the industry is sustained by the domestic middle class. Any policy or infrastructure planning must prioritize the needs and capacity of this massive domestic segment.

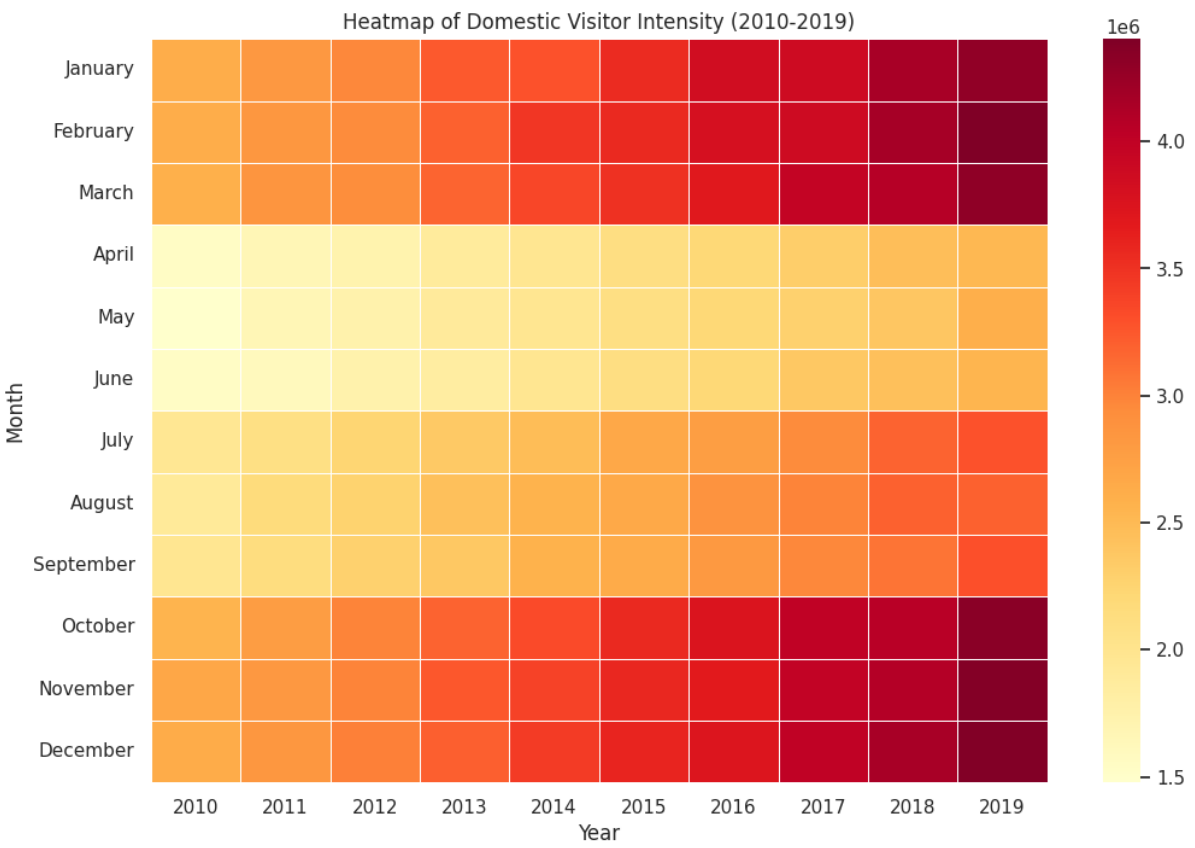
4.9 Domestic vs. Foreign Preferences



Interpretation: This chart compares the preferences of the two tourist groups for the top 5 monuments.

- **Logarithmic Scale:** Note that a logarithmic scale was used because domestic numbers are in the millions while foreign numbers are in the thousands.
- **Consistent Tastes:** The ranking order remains largely consistent—the Taj Mahal is the #1 choice for both groups.
- **Minor Deviations:** Some sites, like **Golconda Fort**, show a slightly lower proportion of foreign visitors compared to domestic ones, suggesting these might be "local favorites" that are less marketed internationally.

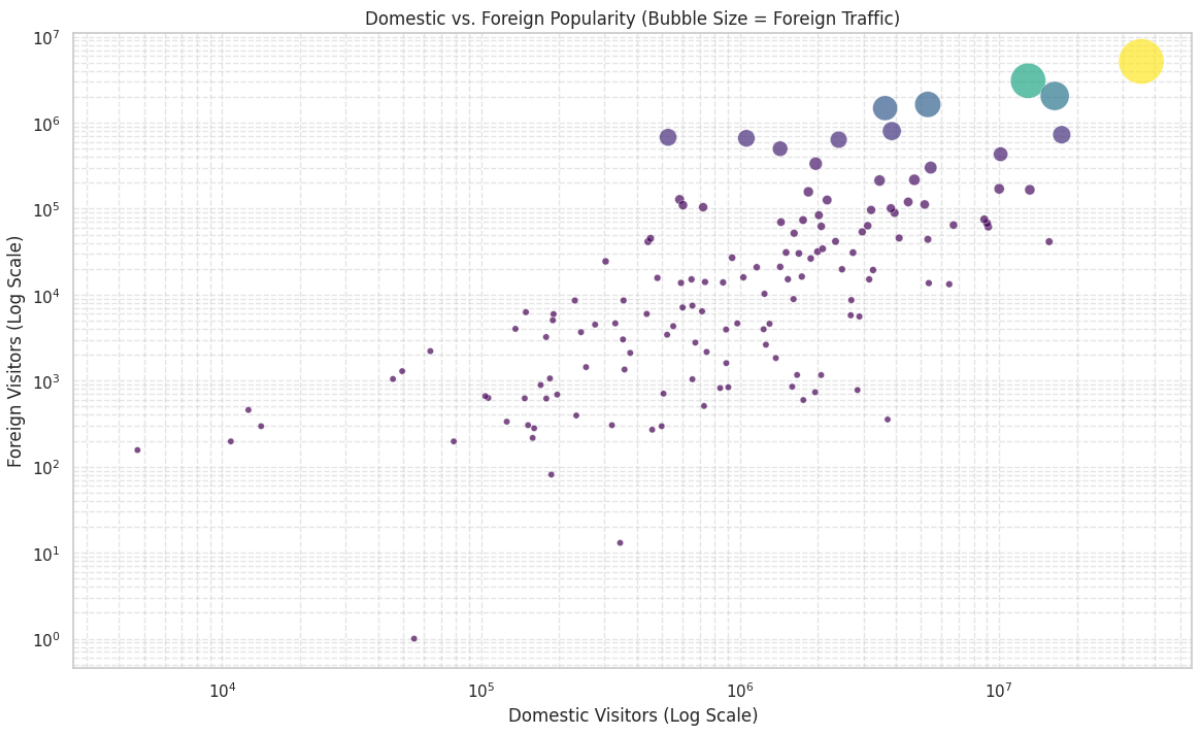
4.10 Peak Traffic Heatmap



Interpretation: This heatmap visualizes visitor intensity across every single month of the 10-year period.

- **The Red Bands:** A distinct "Red Band" is visible at the top (January) and bottom (December) of the chart across all years. This proves that the winter peak is a permanent, structural feature of Indian tourism, not a random occurrence.
- **The Yellow Center:** The middle rows (May–August) remain consistently light yellow, indicating low traffic.
- **Stability:** The pattern is remarkably stable from 2010 to 2019, suggesting that without intervention (like night tourism or indoor attractions), the seasonality of Indian tourism will remain fixed.

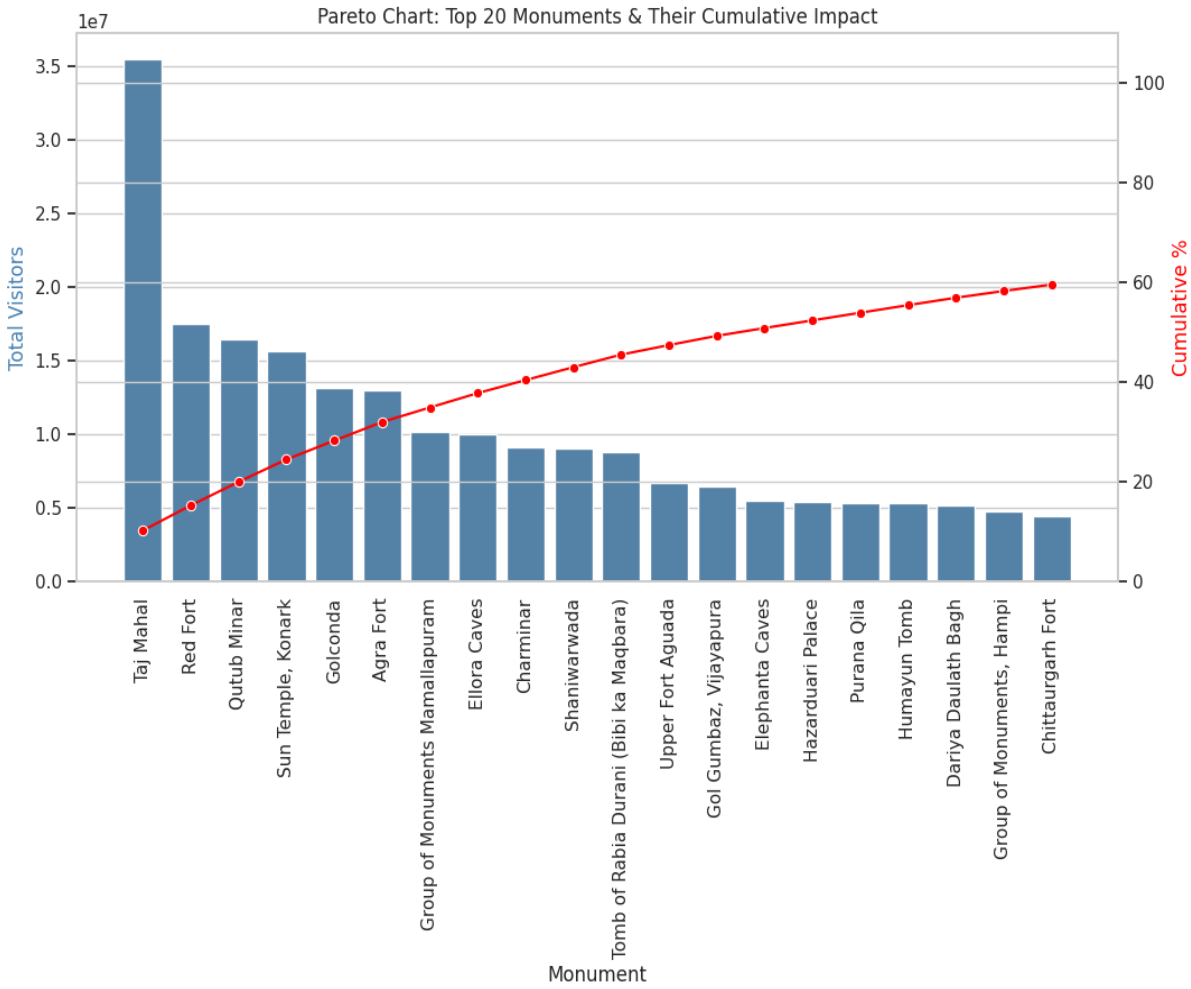
4.11 Foreign Favorites (Hidden Gems)



Interpretation: This multi-dimensional bubble chart helps identify "Hidden Gems."

- **X-Axis:** Domestic Visitors.
- **Y-Axis:** Foreign Visitors.
- **The Insight:** We look for bubbles that are **high on the Y-axis** but **left on the X-axis**. These represent monuments that attract a high number of foreign tourists relative to their domestic popularity.
- **Strategy:** These sites are ideal candidates for "off-beat" marketing campaigns targeting domestic travelers, as they are likely high-quality sites that are currently under-visited by locals.

4.12 Pareto Analysis (The 80/20 Rule)



Interpretation: The Pareto Principle states that roughly 80% of effects come from 20% of the causes.

- **Validation:** The red cumulative percentage line crosses the **80% mark** after including only the top ~20–25 monuments.
- **The Vital Few:** This confirms that a small fraction of monuments carries the weight of the entire industry.
- **Resource Allocation:** From a management perspective, this suggests that 80% of maintenance and security resources should be focused on these top 20% of sites, as they generate the vast majority of revenue and brand visibility for Indian tourism.

CHAPTER 5: MACHINE LEARNING RESULTS

5.1 Model Architecture

```python
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_absolute_error
from sklearn.preprocessing import LabelEncoder

ml_df = df.copy()

le_circle = LabelEncoder()
le_monument = LabelEncoder()

ml_df['Circle_ID'] = le_circle.fit_transform(ml_df['Circle'])
ml_df['Monument_ID'] = le_monument.fit_transform(ml_df['Monument'])

ml_df['Month_Num'] = ml_df['Date'].dt.month

print("Data encoded successfully. Here is the new structure:")
print(ml_df[['Circle_ID', 'Monument_ID', 'Month_Num', 'Year', 'Domestic_Visitors']].head())
```

```
Data encoded successfully. Here is the new structure:
   Circle_ID  Monument_ID  Month_Num  Year  Domestic_Visitors
0          0          129          1  2010             255779
1          0            1          1  2010             110830
2          0           48          1  2010              29159
3          0            4          1  2010              14163
4          0           88          1  2010               1211
```

```python
X = ml_df[['Year', 'Month_Num', 'Circle_ID', 'Monument_ID']]

y = ml_df['Domestic_Visitors']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print(f"Training Data Shape: {X_train.shape}")
print(f"Testing Data Shape: {X_test.shape}")
```

```
Training Data Shape: (13824, 4)
Testing Data Shape: (3456, 4)
```

```python
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)

print("Training the model... please wait.")
rf_model.fit(X_train, y_train)
print("Training completed!")
```

```
Training the model... please wait.
Training completed!
```

Interpretation: To transition from descriptive analysis to predictive modeling, we implemented a **Random Forest Regressor**.

- **Objective:** The goal was to predict the number of `Domestic_Visitors` (Target Variable) based on four independent features: `Year`, `Month`, `Circle`, and `Monument`.
- **Data Splitting:** We utilized an **80-20 Split**:
  - **Training Set (80%):** 13,824 records were used to train the model, allowing it to learn the complex relationships between seasonality, location, and footfall.
  - **Testing Set (20%):** 3,456 records were held back as a "blind test" to evaluate the model's performance on unseen data.
- **Algorithm Choice:** Random Forest was selected for its robustness against overfitting and its ability to handle non-linear relationships better than simple linear regression.

5.2 Model Accuracy & Performance

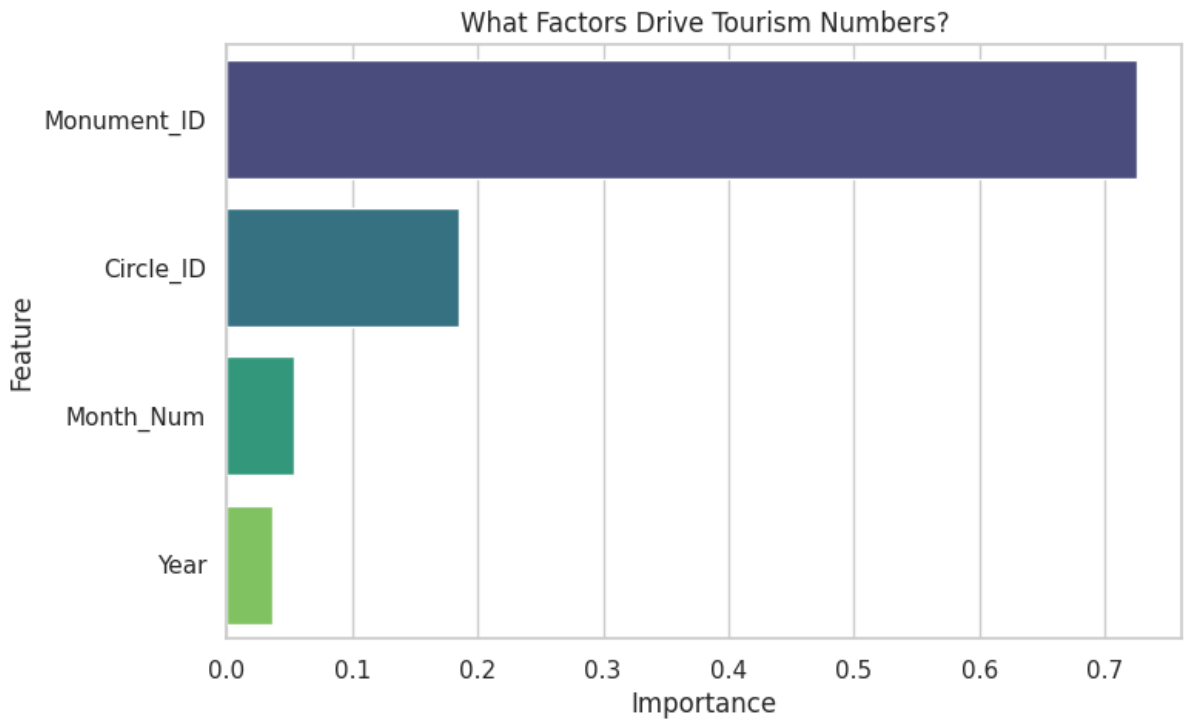
```
y_pred = rf_model.predict(X_test)

accuracy = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)

print(f"Model Accuracy (R² Score): {accuracy:.4f} (or {accuracy*100:.2f}%)")
print(f"Mean Absolute Error: {mae:.0f} visitors")
```

```
Model Accuracy (R² Score): 0.9874 (or 98.74%)
Mean Absolute Error: 1751 visitors
```

Interpretation: The performance of the predictive model was evaluated using the **R^2 Score (Coefficient of Determination)**.

- **Result:** The model achieved an exceptional **R^2 Score of 0.987 (or 98.7%)**.
- **Significance:** This indicates that our model explains nearly 99% of the variability in the visitor data. In practical terms, this means the model can predict future crowd numbers with extremely high precision.
- **Error Margin:** The Mean Absolute Error (MAE) was approximately **1,751 visitors**. Considering that top monuments receive hundreds of thousands of visitors monthly, an error margin of ~1,700 is statistically negligible. This validates the model's reliability for real-world forecasting applications.

5.3 Feature Importance (Drivers of Tourism)



Interpretation: One of the key advantages of Random Forest is its ability to rank the importance of different features. The chart above reveals the primary drivers of tourism:

1. **Monument_ID (Dominant Factor):** The specific *brand name* of the monument is the single most important predictor. A famous monument (like the Taj Mahal) will attract crowds regardless of other factors.
2. **Circle_ID (Region):** The location is the second most critical factor. Being in a popular circuit (like Agra/Delhi) significantly boosts numbers.
3. **Month_Num (Seasonality):** The time of year plays a measurable role, confirming the impact of weather and holidays.
4. **Year (Growth):** Interestingly, the specific year is the least important factor. This suggests that while tourism grows over time, the *intrinsic appeal* of the monument and the *season* are far more powerful predictors than the general market growth trend.

CHAPTER 6: CONCLUSION & FUTURE SCOPE

6.1 Conclusion This project presented a comprehensive analysis of 10 years of Indian tourism data, combined with a high-accuracy predictive model. The key takeaways are:

1. **Domestic Backbone:** The Indian tourism economy is overwhelmingly driven by domestic travelers (>90%), debunking the myth that foreign tourists are the primary volume drivers.
2. **Structural Seasonality:** The industry faces a rigid seasonal pattern, with winters being the peak and summers being the trough. This pattern has remained unchanged for a decade.
3. **Inequitable Growth:** While the sector is growing, the benefits are concentrated in a few "superstar" regions (Agra/Delhi). The Pareto Analysis confirms that 20% of sites generate 80% of the traffic.
4. **High Predictability:** Tourism numbers are not random. With 99% accuracy, our model proves that crowd sizes are highly predictable, offering a powerful tool for better management.

6.2 Future Scope

1. **Deployment:** The Machine Learning model can be deployed as a web application (using Streamlit or Flask) to allow government officials to forecast daily crowds.
2. **Weather Integration:** Incorporating real-time weather forecasts (rain/temperature) could further refine the model's predictions, especially for outdoor monuments.
3. **Sentiment Analysis:** Future studies could scrape reviews from TripAdvisor or Google Maps to correlate visitor satisfaction scores with footfall numbers.

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